

Rahnuma

रहनुमा رہنما

A working guide to the ILM, PANINI and Zistgah stack, from bytes and scripts to compilers, Pāṇini and quantum substrates

प्रकृति

	written	Devanagari hub	Romenagri	and back
Devanagari	प्रकृति	→ प्रकृति	→ prak_ri_ti	→ प्रकृति
Telugu	ప్రకృతి	→ प्रकृति	→ prak_ri_ti	→ ప్రకృతి
Bengali	প্রকৃতি	→ प्रकृति	→ prak_ri_ti	→ the same identity
Urdu, as usually written	ہندوی	→ हनदवी	→ _hana_dawee	→ हनदवी
Urdu, with zer and sukun	ہندوی	→ हिन्दवी	→ _hin_dawee	→ हिन्दवी

One identity, many scripts. Telugu and Bengali meet Devanagari in one hub, become the same Romenagri word, and come back exactly. Urdu meets the hub at what is written: without its short-vowel marks the word reads hanadawee; with zer and sukun it is hindawi. Every cell here was produced by the Hindawi tools while this guide was built.

Abhishek Choudhary, AyeAI. Version 2.0.0, 3 October 2026.

Rahnuma, the edition for the field: Business, administration and law

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Part I: Orientation

1. How to use this guide

In plain words. This guide explains a large body of work on computers, languages and intelligence. You can read it, run the programs on your own computer, or ask an AI to help you, and every example in it was really run.

This guide is for anyone who wants to use, check or extend the work gathered in the *zistgah*, *project-ilm* and *hindawiai* organisations on GitHub: students preparing for GATE, engineers who write compilers, firmware or chip layouts, linguists who work with scripts and sound, and researchers in quantum computing and AI. It asks for curiosity and a laptop, and nothing else.

There are four ways in.

- **Understand.** Part II gives the whole architecture first: where the work comes from, how the parts compose, each term with its nearest analogue and the difference, and the three Acts.
- **Read.** Parts IV and V explain the ideas the work rests on, with the mathematics written out: how numbers and letters sit in memory, what makes a transliteration reversible, how Pāṇini's grammar computes, how a compiler turns text into machine instructions, and how an oscillator network and a quantum circuit attack the same problem.
- **Run.** Part III gives the exact commands to set up Linux, Windows with WSL, or macOS, and to run every component of the estate on your own machine.
- **Ask.** Chapter 24 shows how to put any AI assistant to work on these repositories, and how to check what it tells you.

Conventions used throughout:

- Every example that could run while this guide was being built did run, and the text shows its real output, marked *Executed while this guide was built*. The package's own `gate`, `bash ops/verify.sh`, runs those examples again and fails if any output differs from what is printed here.
- Status words are used exactly. **Tested:** a test in the repository asserts it. **Executed:** it ran and its output is shown. **Written:** the code exists but has not been run here. **Mocked:** a stand-in replaces hardware that is not attached. **Planned:** described, not built.
- Everything said about the repositories was retrieved from the repositories themselves on 29 September 2026: their listings on GitHub, their `CITATION.cff` files and their READMEs. Where a repository does not record something, this guide says so instead of filling the gap.
- A dagger, †, marks a statement that follows the author's working notes and write-ups as supplied on 29 September and 3 October 2026, rather than the repositories.
- Sanskrit is transliterated in IAST. The spelling is British.

The work lives at github.com/zistgah, github.com/project-ilm and github.com/hindawiai, on the site zistgah.org, and in the Zenodo records listed in Chapter 12.

3. Start here: find your way in

In plain words. Four steps to find your part of the guide: who you are, your board or field, what you want to do, and your path with a PDF made for you.

This guide serves a school student and a senior researcher; a village health worker and an investor. Nobody needs all of it. Four steps find your part, on the site or with this page:

1. **Who you are.** A school student, a college student, a postgraduate or doctoral researcher; a teacher; an academic researcher; a working professional, someone in public service, policy or law, someone in community, NGO or village work, an artist, writer or maker; an industry partner or investor; or an AI agent working for a person.
2. **Your board or your field.** A school student chooses a board, ICSE, ISC, CBSE, IGCSE, IB, a state board or another; and a class. Everyone else chooses a field from the international standard classification of fields of study, ISCED-F 2013, the same one FAKIR's lattice uses. It places each subject where its own scholars would: political science is under social and behavioural sciences, not the humanities; law is under business, administration and law; linguistics is under languages.
3. **What you want to do.** Learn the basics, run the software, study or research, teach, contribute, or adopt, fund or invest.
4. **Your path and your edition.** The site lists your chapters in order, with reading times and the components to try first, and offers a PDF edition made for you. Every edition below is a printed selection from this one guide, so all of them stay in step.

Edition	PDF
School student	role-school.pdf
College student	role-college.pdf
Postgraduate or doctoral researcher	role-pg.pdf
Teacher or lecturer	role-teacher.pdf
Academic researcher	role-researcher.pdf
Working professional or engineer	role-professional.pdf
Public service, policy or law	role-public.pdf
Community, NGO or village work	role-community.pdf
Artist, writer or maker	role-creator.pdf
Industry partner, investor or funder	role-industry.pdf
Generic programmes and qualifications	field-00.pdf
Education	field-01.pdf
Arts and humanities	field-02.pdf
Social sciences, journalism and information	field-03.pdf

Edition	PDF
Business, administration and law	field-04.pdf
Natural sciences, mathematics and statistics	field-05.pdf
Information and Communication Technologies (ICTs)	field-06.pdf
Engineering, manufacturing and construction	field-07.pdf
Agriculture, forestry, fisheries and veterinary	field-08.pdf
Health and welfare	field-09.pdf
Services	field-10.pdf

The first public version of this guide, written for advanced researchers and senior professionals, stays as it was at [version 1](#).

Part V: Working with the estate

25. Open source, provenance and contributing

In plain words. How to share, copy and improve this work in the open, and how each piece is stamped with its date and a permanent reference.

Git as a data structure

A Git repository is a directed acyclic graph of commits. Each commit names a tree, the hash of the directory's contents, its parent commits, its author and its message, and every object is addressed by the hash of its content. Change one byte anywhere and every hash above it changes, which is why a commit hash pins an exact state of every file. A branch is a movable name for a commit; a merge commit has two parents.

Contributing on GitHub

```
gh repo fork zistgah/paniniq --clone
cd paniniq
git switch -c order-parameter-bounds
bash ops/verify.sh
git commit -am "Test that the order parameter stays within [0, 1]"
git push -u origin order-parameter-bounds
gh pr create --fill
```

Between switching branches and verifying comes your change. `gh repo fork --clone` also adds the original repository as a remote named `upstream`, so `git fetch upstream && git rebase upstream/main` brings your branch up to date. To report a problem instead of fixing it, `gh issue create --repo zistgah/paniniq` opens an issue.

Licences

The GNU GPL, version 3, is copyleft: anyone may use, study, change and share the software, and whoever distributes a changed version must distribute its source under the same terms. Permissive licences such as MIT and Apache 2.0 allow changes to be kept closed; Apache adds an explicit patent licence. Creative Commons Attribution-ShareAlike 4.0 does for text and media what the GPL does for code. Among the zistgah and project-ilm repositories, every licence that GitHub recognises is a version of the GPL; others carry licence files that GitHub does not recognise, or none that it detects. In every case the repository's own licence file, where it has one, is the authority. This guide's text is CC BY-SA 4.0 and its scripts are GPL-3.0-or-later.

White-labelling, and the commons

White-labelling is selling a product made by one party under another's brand. Under a permissive licence a company can take open code, change it and sell it closed; under the GPL it may still sell it, but it must pass on the source and the same freedoms. Contributing upstream is the opposite of white-labelling: the improvement returns to everyone.

The Aṣṭādhyāyī's review tradition is an old form of the same practice. Kātyāyana's vārttikas proposed amendments to specific rules, Patañjali's Mahābhāṣya discussed and adjudicated them, and the rules were kept intact alongside the commentary, the way a pull request and its review thread are kept alongside the code.

DOIs

A DOI is a persistent identifier that doi.org resolves to a landing page. Zenodo, run by CERN, issues DOIs through DataCite and keeps two kinds: a concept DOI that always points to the latest version, and a version DOI for each release. project-ilm/misty-doi writes the metadata and makes the deposit; its commands are in Chapter 14.

Timestamps with OpenTimestamps

A timestamp proves that a file existed before a certain time without revealing the file. The file's SHA-256 hash is one leaf among many; pairs of hashes are hashed together up a Merkle tree,

$$h_{\text{parent}} = \text{SHA256}(h_{\text{left}} \parallel h_{\text{right}}),$$

and the root is written into a Bitcoin transaction. The proof for one file is the path of sibling hashes from its leaf to the root, $\lceil \log_2 N \rceil$ of them for N files, together with the block that contains the root. Anyone can recompute the path and check that the file existed before that block was mined. The commands are `ots stamp FILE`, then `ots upgrade FILE.ots` once the block is confirmed, and `ots verify FILE.ots`.

Attestations

An in-toto statement is a signed JSON document that binds a subject, file names with their digests, to a predicate saying what was done to them: built, tested, reviewed. It travels in a DSSE envelope, and the SLSA framework defines levels of supply-chain assurance built from such statements. Candor records the estate's typed intents in this form, under each repository's `attest/` folder.

How the estate releases

Every repository carries a `CONTRACT.md` of numbered clauses and an `ops/verify.sh` that checks them and reports PASS, FAIL or UNJUDGED for each. A push happens only when the gate is green, and it writes a receipt, a timestamp, a manifest of hashes in `MANIFEST.sha256` and a seal; a mint then records the DOI in `CITATION.cff` and in the README. The governance repository's contract sets twelve clauses for all of this: sole authorship and AyeAI as the only affiliation; the copyright line; no fabrication; verification by execution; reading before writing; typed gates before anything irreversible, whether a mint, a push, a deletion, an issue or a pull request; staying inside the working folder; provenance travelling with every publication; fork, pull request and merge for shared repositories; small reviewable increments; an agreed split of roles between AIs; and declared repository classes. Clauses C31 to C37 add retrieval before assumption, no vendor lock-in, experience that scales with the user, certification in isolation and in integration, context that travels with every surface, routing as a preference rather than a property, and never bluffing. This package follows the same pattern; its contract is in `CONTRACT.md` and its gate is `bash ops/verify.sh`.

29. The roadmap, for every collaborator and every domain

In plain words. The plan to finish the releases and open the work to everyone: what happens when, which lane each kind of person can join, and every field of work, from farming to medicine to software.

The plan to complete the releases and to open the estate to collaborators is kept as data and run as a GitHub project: one board, an epic for each phase, and an issue for every piece of work, each with its finding, its task, the exact gate that proves it done, the evidence to attach, and what it depends on. The same data gives every collaborator a lane and every domain a lane, so anyone can find where they fit and what is open to them. The roadmap page filters it all by phase, by collaborator lane, by domain family and by owner; at zistgah.org/roadmap.

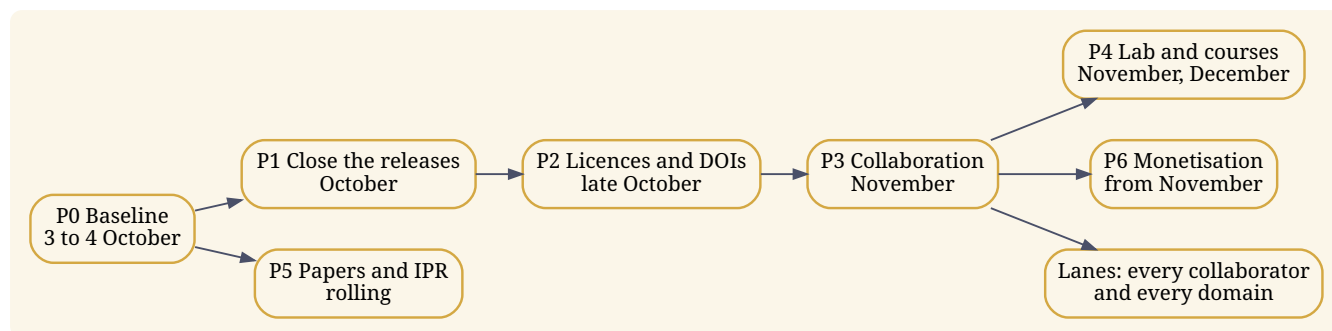


Figure 12. The plan's phases; the lanes run alongside from P3.

Phases

Phase	Window	Goal	Issues
P0 Baseline	Weekend of 3 to 4 October 2026	One measured inventory, the board, tokens with the right scopes, and the list of rulings.	4
P1 Close the releases	10 to 25 October 2026	Everything built is released, and every defect found by execution is fixed.	19
P2 Licences and DOIs	24 October to 8 November 2026	Every repository carries a licence file and correct metadata; minting covers the estate by family.	5
P3 Collaboration infrastructure	31 October to 15 November 2026	A newcomer can go from Rahnuma to a merged pull request in one sitting.	7
P4 Reference lab and courses	November to December 2026	The reference lab runs from launch scripts, with GATE-aligned modules and a pilot.	5

Phase	Window	Goal	Issues
P5 Papers and IPR	Rolling, from October 2026	Disclosure ledger first, then preprints of the four papers with results.	7
P6 Monetisation	From November 2026	Two offers defined and priced, on the dossier standard.	3
Lanes: every collaborator and every domain	Continuous, from October 2026	Every collaborator type and every domain family has a visible lane on the board, with an entry point, open work and a way to propose more.	62

Collaborator lanes

Lane	Start here	Contribute	Issues
School students, 10 to 17	the Rahnuma view for school students; CHAKRA, PANINI's studio, GENIE and the FAKIR dome	learning paths, translations of plain-words summaries, questions	1
Undergraduates: BE, BTech, BSc, BS	the GATE view; Running every component	good first issues, component fixes, capstones seeded from FAKIR	52
Postgraduates and doctoral researchers	the researcher views; Appendices C and D	theses and papers on the open questions, measured results, reviews	52
Teachers, lecturers and lab builders	the teacher view; the reference lab	course modules, lab sheets, pilots, reviews of plain-words text	9
Software engineers	the engineer view; the compilers chapter	code, tests, gates, tooling, pull requests	22
Hardware, embedded and chip engineers	Below the software; zamin, pratik_core_mvp, jugaad28	SPICE, layouts, firmware, device control, ESP32 work	8
Linguists, translators and language workers	the linguist view; writing systems and Pāṇini	language tables, corpora, transliteration rules, measurements	7
Researchers in AI, quantum and cognition	the researcher view; PANINIQ; PEDLER	models, experiments, validation, papers	7
Researchers in the humanities and social sciences	the humanities view; Appendix C	studies of agency, language, knowledge and community	4
Clinicians, health workers and medical researchers	VIKRAM's health section; CEMb and QNBCI	clinical questions, partnerships, review, health-domain tasks	3

Lane	Start here	Contribute	Issues
Community workers, NGOs and village institutions	GramSheel and Project VIKRAM	local needs, pilots, language and field knowledge	30
Farmers, artisans and local enterprises	FAKIR's domain lanes; VIKRAM's livelihood and agriculture sections	domain knowledge, problems worth solving, field testing	24
Artists, writers, filmmakers and musicians	the maker view; the cyclers	works made with the cyclers, recorded as method	3
Makers, roboticists and manufacturers	PANINIphy, pench, transeg	physical realizations, parts, embodiment	6
Policy makers, administrators and public bodies	Appendices C and E; sovereignty in Chapter 19	policy questions, adoption, public infrastructure	5
Lawyers, IPR and standards professionals	provenance and licences; the disclosure ledger	licence review, IPR clearance, standards work	14
Industry partners and companies	Appendix E's adoption path; the component cards	integration, pilots, paid offers, support	9
Funders, sponsors and investors	this roadmap and its measures	funding of lanes, labs and pilots	4
Institutions: colleges, universities and labs	the reference lab and the GATE modules	pilots, courses, joint research	13
AI agents working under a human	any issue's URL; Chapter 22, how to use any AI	issue work through pull requests, with gate output attached	65

Domain families

From FAKIR's lattice: [zistgah/fakir data/lattice.json](#) (ISIC Rev.4, ISCO-08, ISCED-F 2013), 1,505 nodes.

Family	Name	Domains within	Issues
ISIC A	Agriculture, forestry and fishing	54	1
ISIC B	Mining and quarrying	29	1
ISIC C	Manufacturing	232	4
ISIC D	Electricity, gas, steam and air conditioning supply	7	1
ISIC E	Water supply; sewerage, waste management and remediation activities	18	1
ISIC F	Construction	22	1
ISIC G	Wholesale and retail trade; repair of motor vehicles and motorcycles	66	1

Family	Name	Domains within	Issues
ISIC H	Transportation and storage	36	1
ISIC I	Accommodation and food service activities	15	1
ISIC J	Information and communication	42	22
ISIC K	Financial and insurance activities	31	1
ISIC L	Real estate activities	5	1
ISIC M	Professional, scientific and technical activities	35	11
ISIC N	Administrative and support service activities	51	1
ISIC O	Public administration and defence; compulsory social security	11	1
ISIC P	Education	14	7
ISIC Q	Human health and social work activities	21	1
ISIC R	Arts, entertainment and recreation	19	1
ISIC S	Other service activities	26	1
ISIC T	Activities of households as employers; undifferentiated goods- and services-producing activities of households for own use	8	1
ISIC U	Activities of extraterritorial organizations and bodies	3	1
ISCO 1	Managers	46	1
ISCO 2	Professionals	125	1
ISCO 3	Technicians and Associate Professionals	109	1
ISCO 4	Clerical Support Workers	41	1
ISCO 5	Service and Sales Workers	57	1
ISCO 6	Skilled Agricultural, Forestry and Fishery Workers	30	1

Family	Name	Domains within	Issues
ISCO 7	Craft and Related Trades Workers	85	1
ISCO 8	Plant and Machine Operators, and Assemblers	57	1
ISCO 9	Elementary Occupations	50	1
ISCO 0	Armed Forces Occupations	9	1
ISCED 00	Generic programmes and qualifications	6	1
ISCED 01	Education	5	4
ISCED 02	Arts and humanities	13	5
ISCED 03	Social sciences, journalism and information	8	5
ISCED 04	Business, administration and law	10	7
ISCED 05	Natural sciences, mathematics and statistics	13	4
ISCED 06	Information and Communication Technologies (ICTs)	4	16
ISCED 07	Engineering, manufacturing and construction	15	4
ISCED 08	Agriculture, forestry, fisheries and veterinary	9	1
ISCED 09	Health and welfare	12	1
ISCED 10	Services	14	1

The AGI layers are L0 Hardware Definition, L1 Verification, L2 Synthesis/Compilation, L3 Firmware/Embedded, L4 Systems, L5 Parallel, L6 Distributed/Cloud, L7 AI/Deep Learning, L8 Robotics/Embodiment, L9 AGI/ASI. Every domain node, crossed with every layer and every language in ILM's registry, is a possible seed:

$$|S| = |D| \times |L| \times |\Lambda| = 1,505 \times 10 \times 7,867 = 118,398,350.$$

How a collaborator comes in

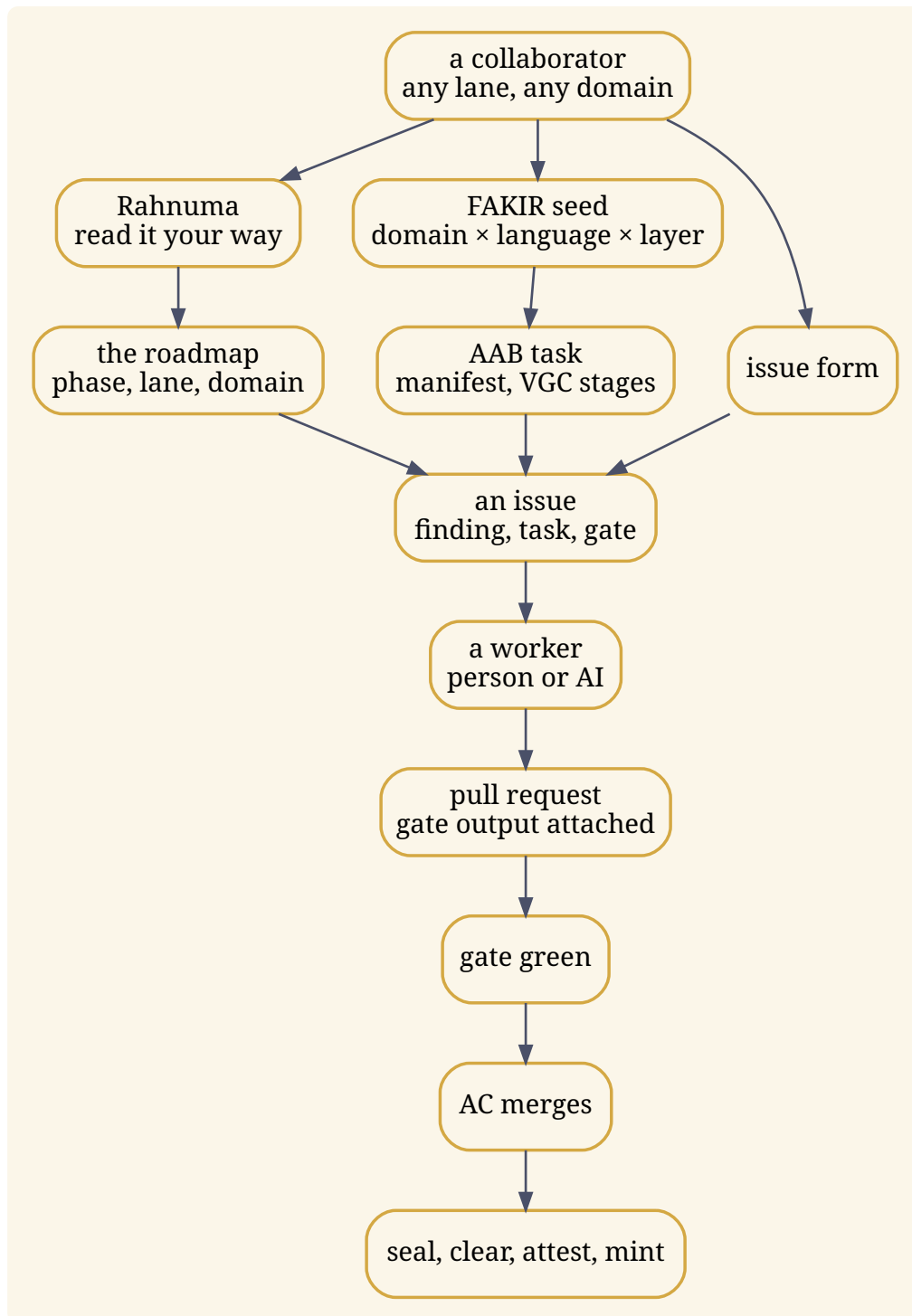


Figure 13. How any collaborator, in any domain, goes from the guide to a sealed contribution.

Four issue forms in [zistgah/governance](#) are the ways in: **Join as a collaborator**, with your lane, domains and languages; **Propose a domain task**, seeded the way FAKIR seeds it, with a domain code, an AGI layer, a language, an intent and a gate; **Report a finding**, with the commands run and what they printed; and **Request a ruling**, for the questions only the author can settle. Every issue says which owner class takes it: a Claude Code session, ChatGPT, Gemini or Fable as workers, or the author himself for push, Pages, mint, merges and rulings.

The work flows the same way for everyone. A worker, a person or an AI, opens a pull request with the gate's output attached; the author merges what is green; anything pushed or minted carries its Candor receipt and goes through seal, clear, attest and mint. Each Saturday the status digest shows every issue as closed, ready, or blocked and by what.

30. What the work would cost: a model-based estimate

In plain words. A measured estimate of how much work the estate holds and what it would cost to rebuild, with and without AI.

How much work does the estate represent, and what would it cost a team to rebuild it, with AI in the process or without? The estimate kit in this guide's repository, under `estimate/`, answers with a recognised model rather than an opinion. It counts every tracked file of every repository once, leaving out vendored, generated, minified and third-party material, measures cyclomatic complexity, and applies COCOMO II.2000, then prices the effort with published salary data and the AI scenario with published token prices and the two controlled trials that bound AI's effect on developer speed. Every rate names its source in `estimate/rates.json`.

Demonstration on 38 repositories cloned for Rahnuma, about a fifth of the estate, most without their full history:

Measure	Value
Unique source lines, each file counted once	243,719
Lines repeated across repositories, counted once	341,227
Words of prose	368,689
Functions measured; above complexity 10; above 20	9,288; 546; 191

Case	Effort, person-months	Schedule, months	Cost, India	Cost, United States
low	701	27.1	₹12.9 crore	\$7.5 million
likely	1,239	35.3	₹41.0 crore	\$18.6 million
high	2,279	47.3	₹130.7 crore	\$49.9 million

AI in the process	Time ratio	Human effort, person-months	AI usage
low	0.44	554	\$140
likely	0.75	944	\$578
high	1.19	1,498	\$5,861

The AI usage costs a few hundred to a few thousand dollars at list prices; the outcome turns on the time ratio, which the trials put between 0.44 and 1.19.

To run it on the whole estate: `bash estimate/estimate.sh --clone`, which clones every repository with full history, or `bash estimate/estimate.sh --root <folder of clones>`. The report also sets the model against the estate as it was actually built, from the distinct days in its git history, with hours per day and AI spend that the author sets in `rates.json`.

COCOMO prices the construction of what is written down. It does not price the research, the invention and the judgement behind it, so every figure here is a floor, not a valuation.

Appendices

C. For researchers in the humanities and social sciences

In plain words. A guide for researchers in the humanities and social sciences: the questions each field can ask of this work, and where to find and test the answers.

Each discipline below is given a question the architecture makes concrete, where to look, what can be examined or run, and what must be kept distinct. These are research questions, not claims that any discipline has validated the architecture. The technical components specify relations; the disciplines bring the questions through which those relations can be investigated. Neither vocabulary replaces the other: PEDLER is a computational object that psychologists and philosophers can interrogate as a model of agency, not psychology; GramSheel is an infrastructure that sociologists can interrogate as a model of capability distribution, not a development programme; ILM is a computational object that linguists can interrogate as a model of multilingual representation, not linguistics.

Political science, law and governance

The question. Who holds agency and computational authority, and what counts as admissible evidence when machines take part in decisions? **Where to look.** The Proclamation of Individual Equity and recursive sovereignty (Chapter 4); Humanesque and its projections (Chapter 10); PATAL, GramSheel and Project VIKRAM; Natural Justice as a gate in PEDLER; VGC and the provenance order seal, clear, attest, mint (Chapter 25). **What can be examined.** The text of PoIE (zistgah/poie, doi:10.5281/zenodo.21397274); the executable contracts that gate every repository (zistgah/governance), each a set of clauses a machine checks; the Candor receipts, which record signed intent as in-toto statements. **Keep distinct.** PATAL is not democracy, and Humanesque is not political theory; Natural Justice in PEDLER is a gate on the passage from intent to Act, not a general ethics.

Sociology and development studies

The question. Can technological capability be distributed through communities rather than concentrated in distant institutions, and can villages produce knowledge and technology rather than only consume it? **Where to look.** GramSheel and its principles, health, education, justice, livelihood and peace; Project VIKRAM and its thirteen sections; FAKIR's lattice of every lawful economic activity under ISIC, ISCO and ISCED; the reference lab. **What can be examined.** VIKRAM's journal and its section pages (pvjournal.github.io), launched on 3 May 2020; FAKIR's explorer, where each point seeds a task for a domain, a language and a layer; the open contribution model of the repositories, through issues and pull requests. **Keep distinct.** GramSheel is the foundation and its principles; VIKRAM is the technical platform; neither is a market segment or a programme of delivery to passive recipients.

Psychology and cognitive science

The question. How does an agent move from a state and an inclination to an action, and how does a machine's projected authority affect what people accept? **Where to look.** PEDLER's loop and its primitives (Appendix B); COPA and its Authority Projection Index (Chapter 8); the Quantum Neuromorphic BCI programme, with its active research and its directions needing clinical partners. **What can be examined.** The PEDLER engine in PANINIq, with its tests (Chapter 23); COPA's index,

which can be computed on any transcript as explanatory tokens before the first empirical action over all tokens. **Keep distinct.** Inclination is not intention, and intention is not an Act; COPA measures projected authority, not accuracy or intent, and a single reading proves nothing about intent.

Linguistics

The question. How are language, script and meaning preserved when linguistic structures become computational representations, and can any language travel in any script? **Where to look.** Writing systems and reversible transliteration (Chapter 19); Pāṇini's grammar as a formal system (Chapter 20); Hindawi and the symbol bridge (Chapter 21); ILM and tajziya. **What can be examined.** Every Romenagri result in this guide can be rerun on your own corpus with the commands of Chapter 15: the Brahmi hub, the round trip, the Urdu projection with and without tashkil. ILM's registry lists 7,867 languages and 226 scripts, and its keyword standards number 13,668, every one a valid ASCII identifier. **Open.** The share of Urdu words whose vowels must come from the language layer, which the author estimates at 15 to 20 per cent, awaits measurement, and the reverse filter into Perso-Arabic script is a first cut. **Keep distinct.** ILM is not linguistics and Romenagri is not romanisation; script, language and standard are three axes.

Anthropology and cultural studies

The question. How do identity, situated knowledge and cultural plurality survive technological transformation? **Where to look.** The three projections, Kaivalyik, Zistgah and Cosmopolis, and the names table (Chapter 10); Jyotish as the Kaivalyik projection of CHAKRA; the second pillar of linguistic equity, any language in any script. **What can be examined.** What the Devanagari hub keeps and loses when Urdu passes through it, printed in Chapter 21; how a component keeps one stem and takes a culturally sensitive name in each projection. **Keep distinct.** A projection is a complete traversal of the whole architecture, not a translation of a master copy; Zistgah's Persian and Islamicate dimension stays explicit.

Philosophy

The question. What constitutes knowledge, action, identity and continuity when the substrate changes? **Where to look.** PEDLER's primitives; VGC; the provenance systems; TransEg; Synthematic Pragmatic Realism; the rule that a record not yet retrieved is not thereby false. **What can be examined.** An applied epistemology already in use: GENIE's nine epistemic tags (definition, axiom, assumption, derivation, conjecture, hypothesis, empirical, open, failed), the four provenance states of a statement (retrieved, inferred, proposed, unresolved), and the status words of this guide (tested, executed, written, mocked, planned). † The write-ups put the underlying problem as an unconstrained system making an epistemic state transition without an admissible reason. **Keep distinct.** SPR is a stated primitive of the architecture; generic metaphysics must not be substituted for it.

Education

The question. Does technology hand out answers, or does it increase human capability? **Where to look.** The reference lab and the GATE chapter (Chapters 16 and 26); the reader views of this guide; the cyclers as recorded, reproducible method; VIDYA, the bridge to AyeAM. **What can be examined.**

The component cards of Chapter 15 as laboratory sheets, each with what to type and what you should see. † The write-ups also name AYE Learn and AYE Sum among the educational work.

History and the history of computing

The question. What was built, when, and how can priority be checked without argument? **Where to look.** The dated record (Chapter 5) and its primary sources: the SourceForge and Savannah projects, the announcement of 19 December 2005, and the Zenodo records. For context rather than lineage, Gandhi's *Constructive Programme* (1941) placed village industries, education and the provincial and national languages inside the work of self-rule; Chapter 10 sets out where the estate's questions meet it.

Economics and political economy

The question. Can advanced capability become locally reproducible and sustainable rather than externally concentrated? **Where to look.** FAKIR's ISIC lattice as a map of economic activity; the open licences of the repositories (Chapter 25); the reference lab's cost target. † The notes treat monetisation, through products, services, courses, labs and licensing, as the mechanism that sustains the open work.

E. For practitioners: teachers, policy makers, communities, makers and engineers

In plain words. A guide for teachers, policy makers, community workers, artists and engineers: where to start and how to take part.

Teachers and lab builders. Start from the reference lab, the GATE chapter and the component cards, which double as laboratory sheets: each says what to install, what to type and what a student should see (Chapters 16, 26 and 15). The reader views let each student follow a path suited to their age and background, and remember what they have read.

Policy makers and administrators. The questions that matter are in Chapter 10 and Appendix C: sovereignty in its testable technical sense, the ability to rebuild, verify, modify and run (Chapter 21); provenance that makes claims checkable; and infrastructure that leaves capability with communities. The executable contracts of the repositories show what verifiable governance of software looks like in practice.

Community workers and health practitioners. Project VIKRAM's sections cover education, health, livelihood, justice, environment and habitat, food and agriculture, and peace; its health section accepts submissions as issues. The Urdu edition and the Brahmi hub show work in people's own scripts; CHAKRA and Jyotish run offline.

Artists, writers and makers. The cyclers produce print, film, sound, immersive work, embodied work and the record; matba turns posters into a sealed book; GENIE writes and paints; PANINIphy turns a physical idea into parts. Every cycle is recorded as method, so a piece of work can be repeated, shared and credited.

Engineers in industry. Adopt one component at a time: read its card, clone it, run it, read its `CONTRACT.md` and its gate, check its DOI and provenance, then integrate it through its interface. † The notes give the full adoption ladder: the canonical artifact, a one-page explanation, the nearest industry analogue and the exact difference, a working demonstration, the repository, the DOI and timestamp, the interface, a case study, and the commercial route.

Contributors. Fork, open an issue, send a pull request. Every repository's gate runs on your change; every claim must point to its evidence; the AI you use is welcome, and its report that something passed is not the evidence that it did.

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